Exploratory Data Analysis Forecasting Handoff Report

Evidence selected from the project Jupyter notebook: annual_indices, daily, and city_info. Analysis uses information dated no later than 31 December 2018.

Purpose: document which historical observations are trustworthy, which patterns are visible in the data, and which hypotheses the forecasting team should validate. All charts reproduced below are saved outputs from the supplied notebook.

Executive takeaways

Data reliability. Target-specific 10-day completeness screening is necessary; numerical annual values can be unreliable when many daily observations are absent.

Recent climate change. Across eligible cities, 96.0% show positive recent annual-mean-temperature slopes and 96.4% show positive CDD slopes.

Forecasting signal. Average temperature and CDD have greater detrended lag-1 autocorrelation than annual precipitation; the value of these signals for seven-year prediction remains untested.

Daily verification. All 609 initially uncertain CDD city-years were resolved using daily temperature completeness.

Station history. 180 of 210 cities combine multiple station periods; 19 transitions were flagged for closer investigation, not for automatic correction.

Reading guide: each selected result is followed by "Main inference" (what the data supports) and "Why it matters" (its implication for historical forecasting). Numerical values are descriptive; model efficacy requires out-of-time backtesting.

1 Dataset overview

Dataset	Scale	Role in analysis
annual_indices	28,052 city-years; 210 cities	Annual forecasting targets and climate/time-series EDA
daily	10,245,754 daily rows	Completeness checks, annual-index validation, seasonal/heat diagnostics
city_info	461 station-period records	Threaded station histories and potential discontinuity screening

Main inference. The annual table is the forecasting target dataset; the daily and city_info tables serve mainly as verification and context rather than additional raw rows to merge wholesale.

Why it matters. Forecasting can operate at one row per city and year, joining only the validated quality indicators and any justified historical features by city ID and year.

2 Data quality and missingness — annual_indices

2.1 Target-specific completeness

Annual target	Initially valid city-years	Valid share
Avg Temp	24,583	87.63%
ToT Prcp	24,858	88.61%
Max Tmax	24,868	88.65%
cdd (before daily check)	23,974	85.46%

Main inference. Between roughly 11% and 15% of city-years failed initial target-specific screening. A value can be non-null while still having too many missing daily observations.

Why it matters. Use the target-specific valid/clean columns or flags rather than dropping all rows with any missing field or training blindly on every numeric annual value.

2.2 Missingness changes through history

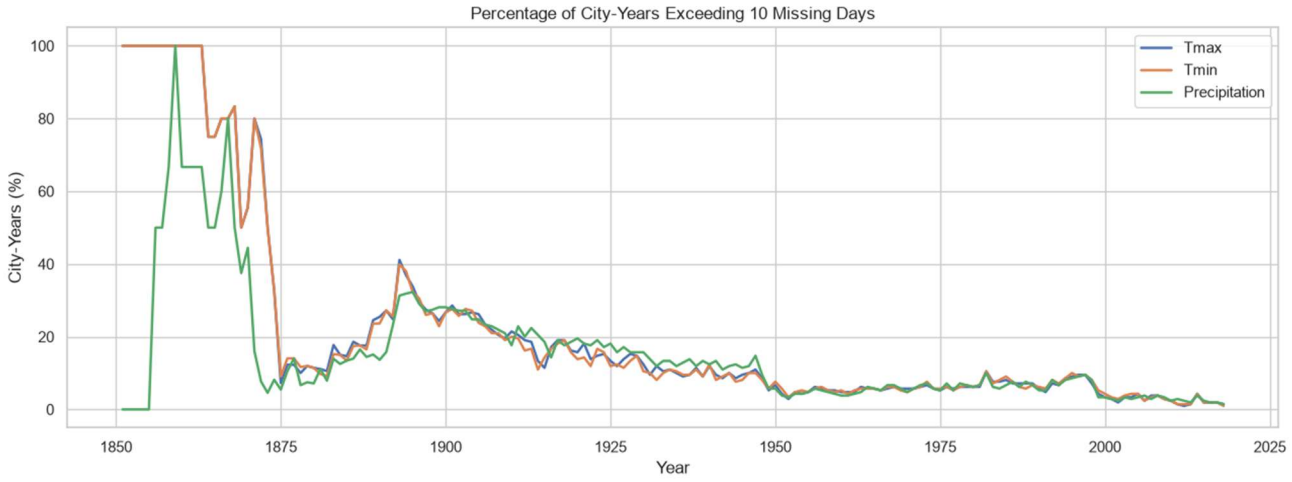


Figure. City-years exceeding the 10-day missingness threshold for each weather measure.



Figure. Valid target coverage across historical periods.

Main inference. Completeness improves strongly over time: average-temperature valid coverage is about 80.6% before 1950 versus 95.2% in 1989–2018. Some individual cities remain poorly observed even recently.

Why it matters. Recent-history windows may offer cleaner targets, but the training pipeline must still handle differing historical coverage by city and target. Do not assume all 210 cities have equally complete recent histories.

3 Target distributions, outliers, and variability

3.1 Distributions of the four targets

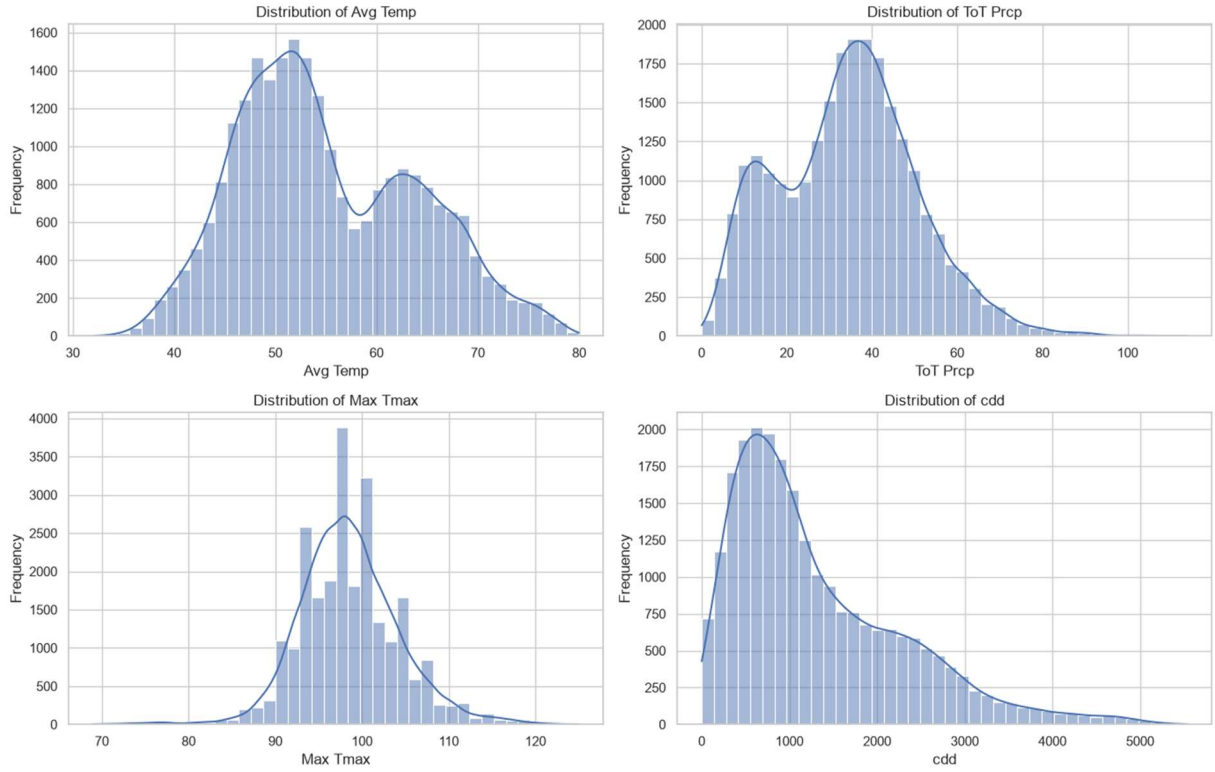


Figure. Quality-filtered annual target distributions across all available city-years.

Main inference. Precipitation and CDD are right-skewed; pooled temperature distributions combine cities with very different baseline climates. The targets are not identically distributed.

Why it matters. Forecast evaluation and candidate transformations should be chosen per target; pooled error metrics should not obscure city or target differences.

3.2 Variation differs by city

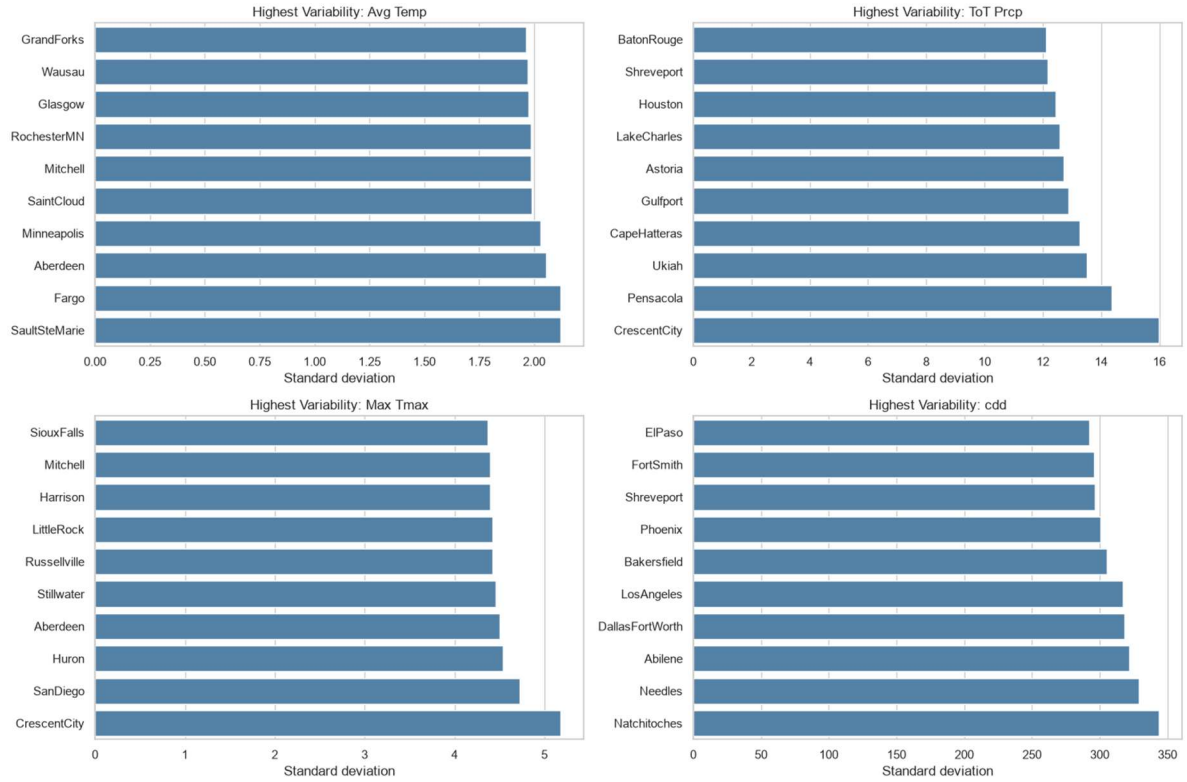


Figure. Cities with the greatest recent variability, by target.

Main inference. Historical year-to-year variability differs widely across cities: northern/interior cities feature prominently for annual temperature, while several coastal/southern cities have high precipitation variability.

Why it matters. Comparing forecasting performance by city and target is important; a single overall accuracy figure could conceal consistently difficult locations.

3.3 Statistical outliers are not necessarily errors

Target	Flagged city-years	Share of valid observations
Avg Temp	298	1.21%
ToT Prcp	355	1.43%
Max Tmax	299	1.20%
cdd	318	1.33%

Main inference. Only a small share of quality-filtered observations were flagged by the exploratory outlier rule. Some extremes occurred with complete daily records.

Why it matters. Keep extreme observations unless there is evidence of measurement error; extreme outcomes are part of the forecasting problem, particularly for hottest-day temperature.

4 Historical climate trends

4.1 Long-term versus recent slopes

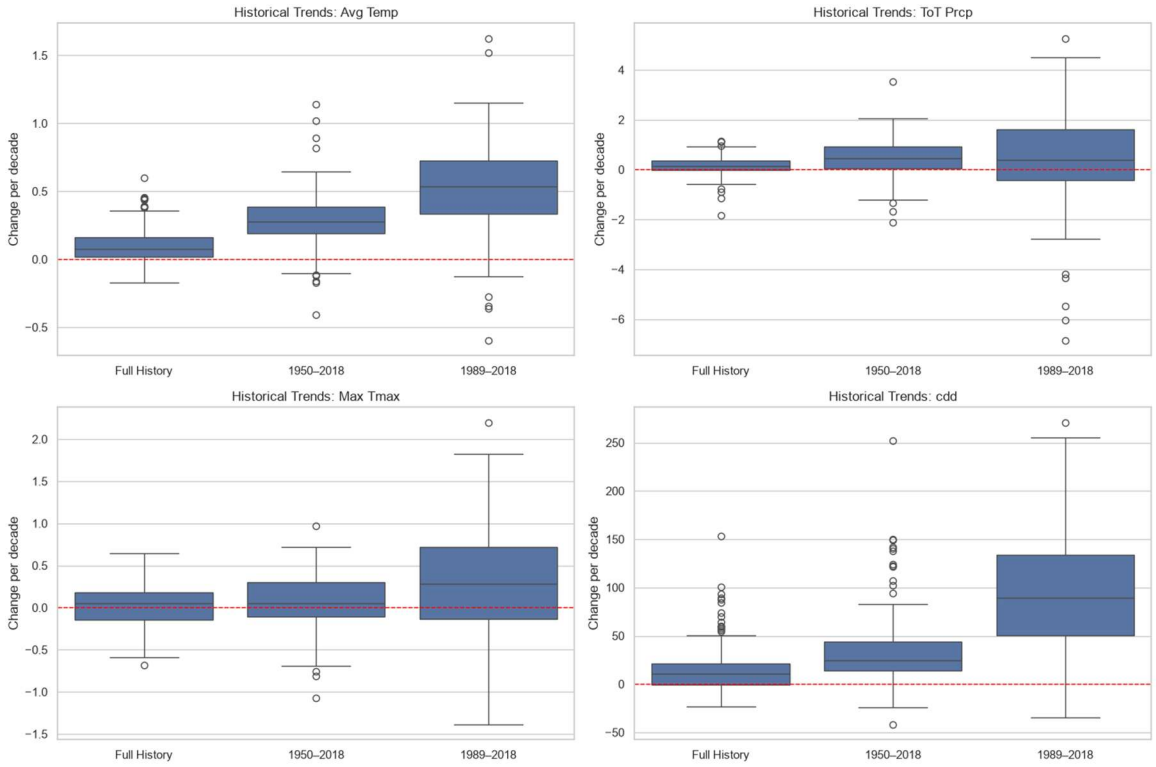


Figure. Distribution of city trend slopes over full history, 1950–2018, and 1989–2018.

Target	Full-history median trend	1989–2018 median trend
Avg Temp	+0.076 °F/decade	+0.538 °F/decade
Max Tmax	+0.047 °F/decade	+0.287 °F/decade
ToT Prcp	+0.140 in/decade	+0.393 in/decade
cdd	+10.64 °F-days/decade	+89.46 °F-days/decade

Main inference. Recent median warming and rising cooling demand are much stronger than their full-record slopes. Precipitation slopes are more heterogeneous and noisier.

Why it matters. Recent-window and full-history trend extrapolations should be compared in time-based validation instead of assuming either window is always best.

4.2 Widespread increases do not imply strong annual predictability

Target (1989–2018)	Cities with positive slopes	Median trend R^2
Avg Temp	96.0%	0.109
cdd	96.4%	0.162
Max Tmax	69.5%	0.018
ToT Prcp	62.3%	0.021

Main inference. Positive recent slopes are widespread for temperature and CDD, but simple linear trends explain a limited amount of individual year-to-year variation, particularly for precipitation and hottest-day temperature.

Why it matters. Trend features should be tested alongside stable baseline forecasts; trend direction alone is not a reliable forecast of a specific future year.

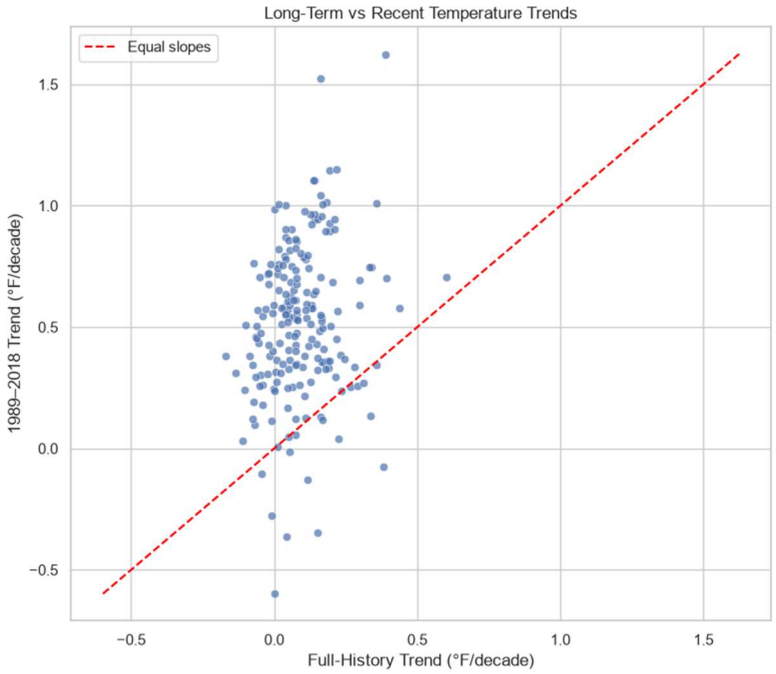


Figure. Full-history versus recent city-specific annual temperature slopes.

4.3 Temperature and CDD move together

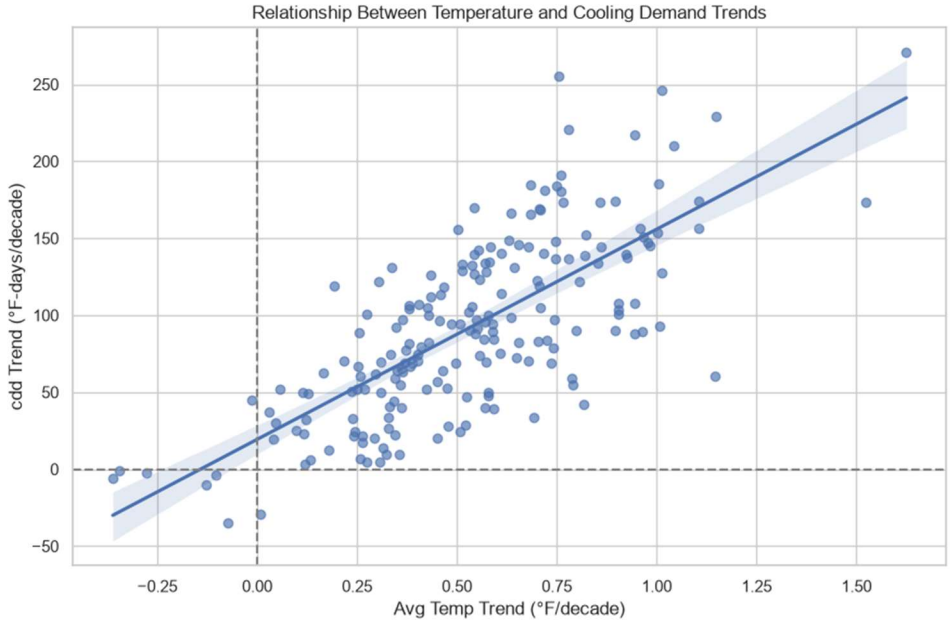


Figure. Relationship between recent annual temperature and cooling-demand trend slopes.

Main inference. Cities with stronger warming slopes generally also show stronger increases in CDD; this is partly expected because CDD is calculated from daily temperatures.

Why it matters. Joint or temperature-informed CDD features are worth testing, but the apparent relationship is not independent evidence of future predictive power.

5 Differences between cities

5.1 Geography and climate baselines

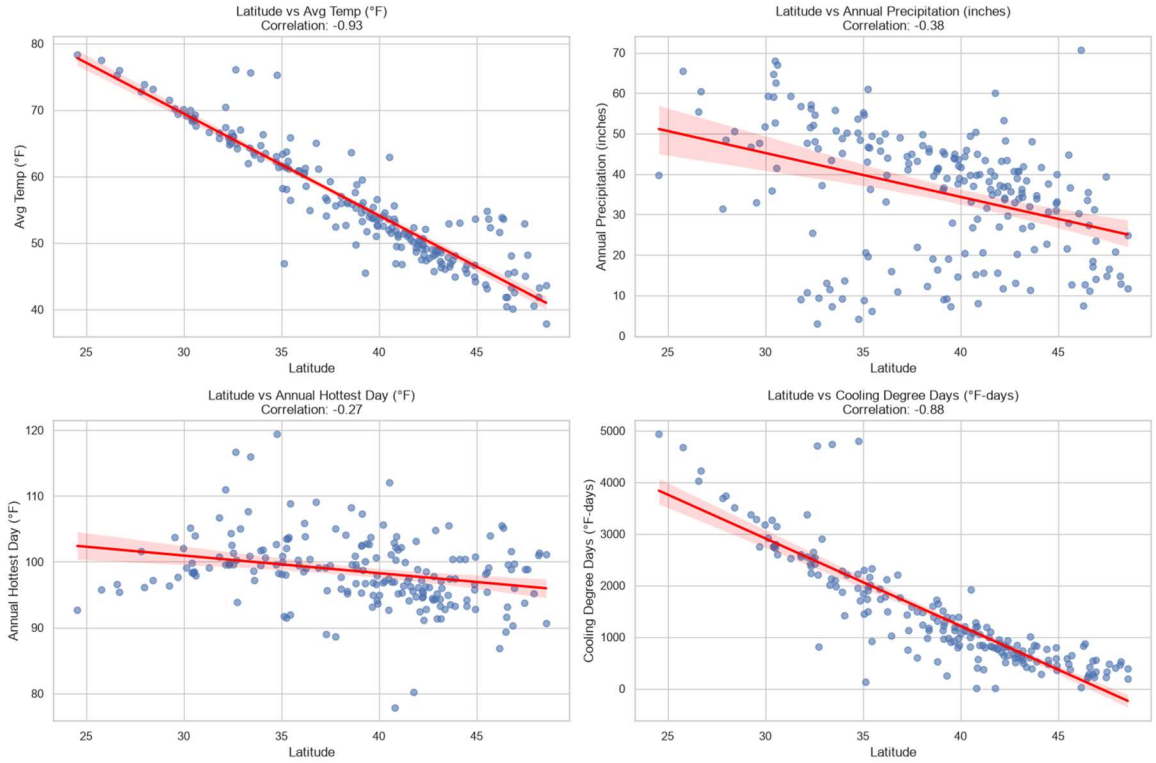


Figure. Latitude compared with each city's 1989–2018 climate averages.

Target	Correlation with latitude
Avg Temp	−0.93
cdd	−0.88
ToT Prcp	−0.38
Max Tmax	−0.27

Main inference. Latitude relates strongly to average temperature and CDD, but much less strongly to precipitation or hottest-day temperature.

Why it matters. City identity, geographic location, and climate baselines are candidate features in a pooled model; latitude alone will not predict annual deviations.

5.2 Baseline climate is not the same as trend

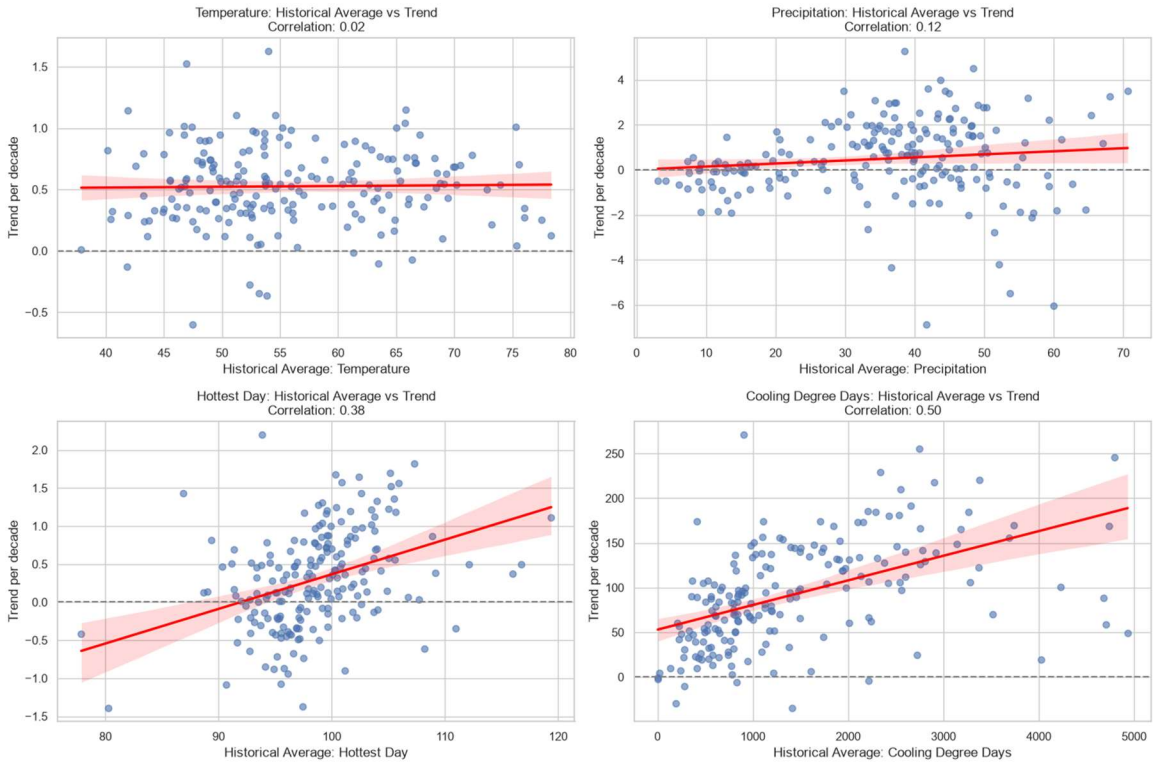


Figure. Historical city averages compared with 1989–2018 trend slopes.

Main inference. The city mean temperature and recent temperature slope are essentially uncorrelated (r≈0.018), while mean CDD and recent CDD slope have a moderate association (r≈0.504).

Why it matters. A pooled model may need both stable city-level baselines and distinct time-trend information.

5.3 Shared annual fluctuations among nearby cities

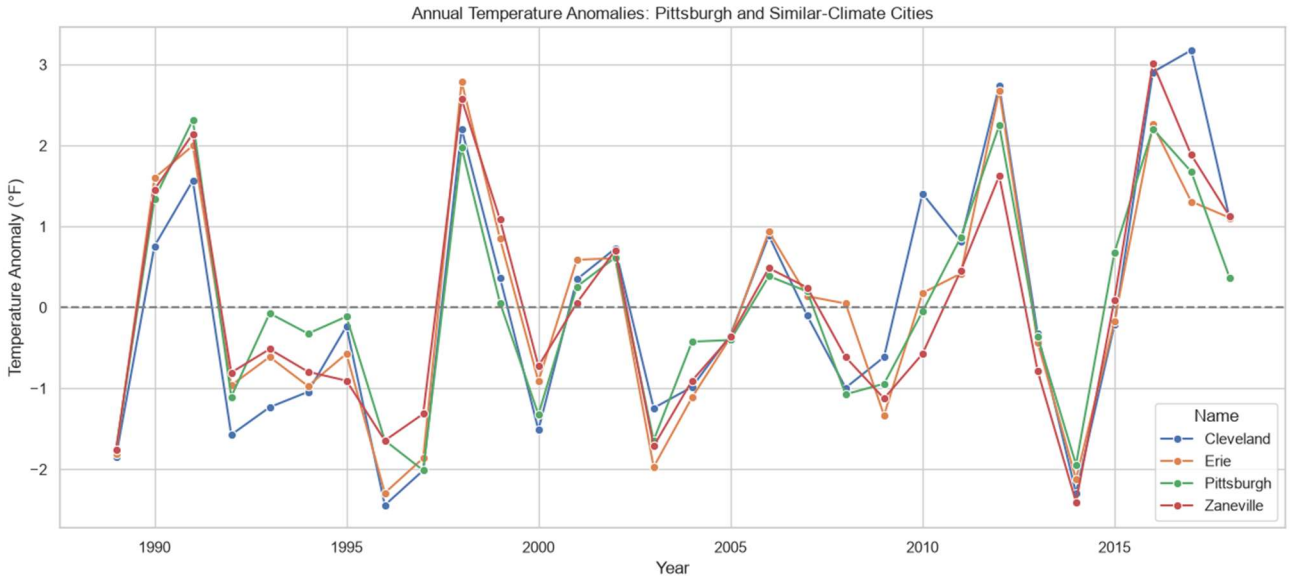


Figure. Pittsburgh, Cleveland, Erie and Zaneville annual temperature anomalies.

Anomaly comparison	Correlation
Pittsburgh ↔ Cleveland	0.92
Pittsburgh ↔ Erie	0.94
Pittsburgh ↔ Zaneville	0.94

Main inference. Some close-by cities exhibit very similar annual departures from their own temperature means; similarity of average climates alone does not guarantee similar warming rates.

Why it matters. Regional or pooled models are reasonable experiments. Future contemporaneous weather in neighboring cities cannot be used unless it is also forecast; shared history is not future information.

6 Annual time-series structure and predictability

6.1 ADF test summary

Target	Eligible city series	ADF "ct" reject unit-root null
Avg Temp	196	71.94%
Max Tmax	197	95.43%
ToT Prcp	198	92.42%
cdd	185	78.38%

Main inference. Most tested series reject the ADF unit-root null under a constant-plus-linear-trend specification, but there is substantial variation across city-target combinations. ADF rejection does not by itself establish stationarity.

Why it matters. Do not impose one differencing or stationarity assumption on all series. Time-series model assumptions should be checked on each fitted training window.

6.2 Autocorrelation before versus after detrending

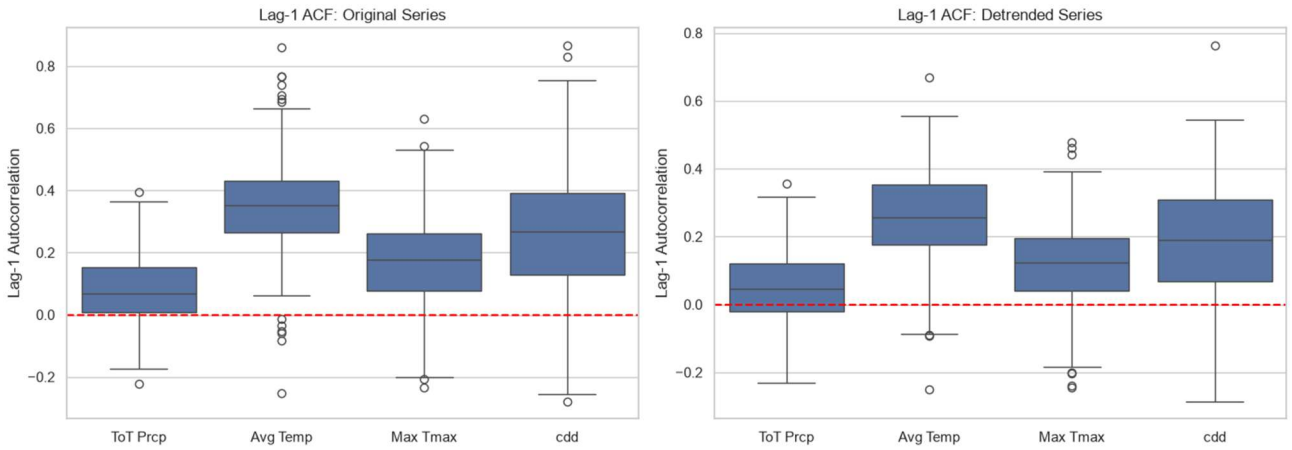


Figure. Across-city distributions of annual lag-1 ACF before and after linear detrending.

Target	Median raw lag-1 ACF	Median detrended lag-1 ACF
Avg Temp	0.351	0.257
cdd	0.267	0.191
Max Tmax	0.177	0.125
ToT Prcp	0.069	0.046

Main inference. Temperature and CDD have more year-to-year persistence than precipitation and Max Tmax. Detrending reduces the ACF, revealing that some apparent persistence reflects slower climatic movement.

Why it matters. Lag predictors may be more promising for temperature and CDD. Seven-year forecasts require direct multi-horizon or recursive design without using actual observations after the cutoff.

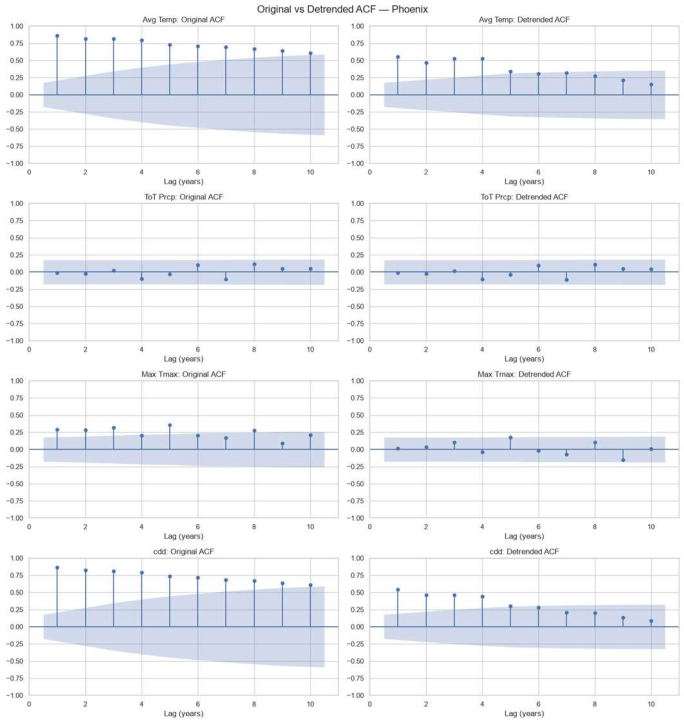


Figure. Original versus detrended ACF for Phoenix (representative city).

7 Daily dataset: validation and additional climate features

7.1 The CDD missingness question is resolved

CDD classification result	City-years
Initially uncertain	609
Resolved valid	261
Resolved invalid	348
Still unresolved	0
Final valid CDD city-years	24,235
Final invalid	2,611
Final missing CDD target	1,206

Main inference. The daily record establishes on which days Tmax and Tmin were both present, resolving all 609 ambiguous annual CDD completeness cases.

Why it matters. The forecasting member should use the updated CDD quality flag (24,235 valid city-years), not the initial annual-only count.

7.2 Agreement between daily-derived and supplied annual values

Annual target	Compared city-years	Mean absolute difference
Avg Temp	26,859	0.0033 °F
ToT Prcp	28,052	0.0000 in
Max Tmax	26,868	0.0000 °F
cdd	26,846	0.0000 °F-days

Main inference. Recomputed daily aggregates agree with the annual targets for nonmissing comparisons, up to tiny temperature rounding differences.

Why it matters. The annual target formulas are consistent with the source daily observations. Agreement does not override missing-data quality thresholds.

7.3 Preliminary seasonal and hot-day diagnostics

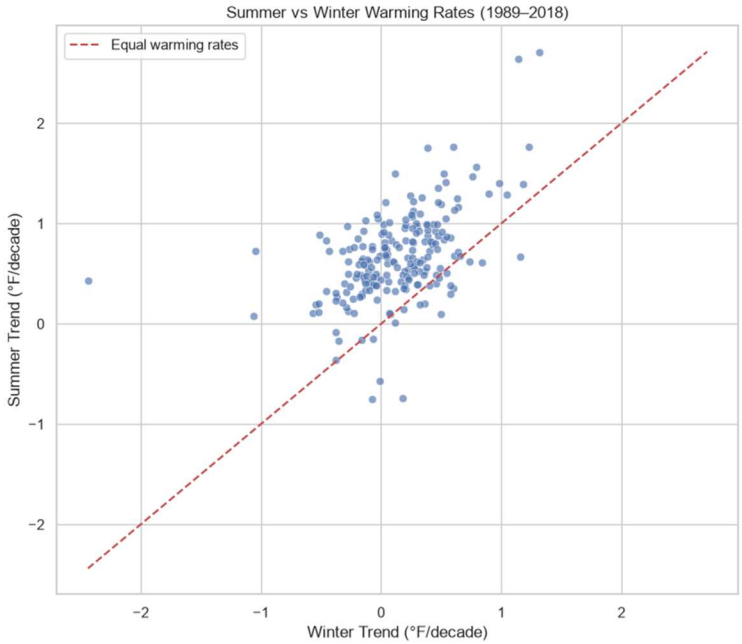


Figure. Summer versus calendar-year winter temperature slopes, 1989–2018 (exploratory).

Season	Median warming slope
Summer	+0.626 °F/decade
Winter	+0.149 °F/decade

Main inference. The exploratory seasonal analysis suggests stronger recent summer warming, but the monthly/seasonal calculations did not fully enforce completeness; the results are provisional.

Why it matters. Quality-checked seasonal warming rates and historical counts of ≥90°F days may be useful additional features for CDD or heat forecasts, but they must be rechecked and backtested before use.

8 City info: stitched weather-station records

8.1 Prevalence of station changes

Station-history measure	Observed
Cities	210
Station-period records	461
Cities with >1 station period	180
Recorded transitions	251
Transitions during 1989–2018	32 (across 30 cities)

Main inference. Most cities have climate records stitched from several contributing station periods. This complicates interpretations of full-history slopes.

Why it matters. Station-transition dates provide quality-context flags and should be considered in sensitivity analyses, not treated as proof that the observations are erroneous.

8.2 Transition anomalies compared with regional changes

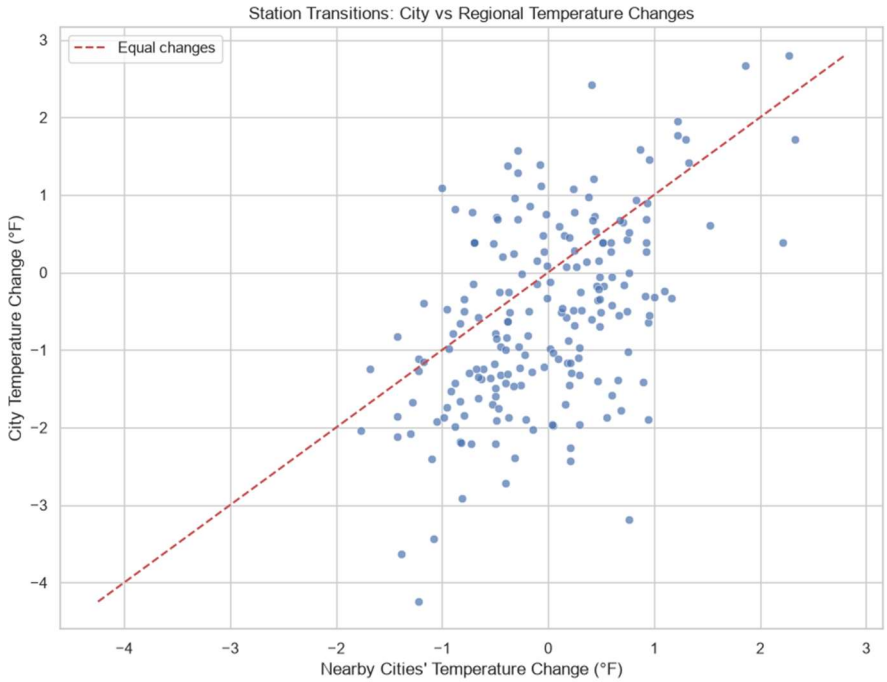


Figure. Five-year pre/post-transition average-temperature change: city versus nearby-city median.

Flagged city / year	City change	Nearby-city change	Difference
Marquette / 1979	−3.19°F	+0.76°F	−3.95°F
Seattle / 1945	−4.25°F	−1.22°F	−3.02°F
Flagstaff / 1950	−1.89°F	+0.95°F	−2.84°F
Houston / 1969	−1.87°F	+0.55°F	−2.42°F

Main inference. Nineteen evaluated transitions were flagged because a city's before/after change differed by at least 2°F from the median change in nearby comparison cities. These flags are investigative, not confirmed station biases.

Why it matters. Test whether trend-based forecasts are sensitive to flagged transition periods; do not delete or "correct" these observations automatically.

9 EDA-to-forecasting handoff decisions

Evidence from EDA	Experiment / handoff for modeling
Target-specific incomplete annual observations	Use separate valid-target flags; define treatment per target and per training fold
Recent warming > full-history slope	Backtest different trend windows against simple historical-mean baselines
Weak precipitation annual-lag persistence	Test precipitation climatology and trend approaches before complex autoregression
Large differences in city baselines	Compare global/pooled models with per-city or region-based baselines
Correlated regional temperature anomalies	Test shared historical features; never use unknown future nearby-city outcomes
Updated daily CDD checks	Use final cdd_clean / cdd_quality classifications
Preliminary seasonal indicators	Recompute with strict monthly completeness before testing as features
Nineteen flagged station transitions	Retain review flags, assess sensitivity, avoid unverified adjustments
Seven-year target horizon	Use rolling-origin hindcasts that produce complete seven-year blocks from each cutoff

Open methodological limitations

1. Station-transition comparisons are screening tests using five-year windows and nearby-city medians; they do not identify causal measurement biases. Comparison cities can have their own discontinuities.

2. Seasonal/monthly aggregates and hot-day diagnostics require stricter completeness screening before they are used in feature engineering.

3. EDA trend, autocorrelation, scaling and similarity estimates use data through 2018. Every historical validation fold must recompute its statistics strictly from information available by that fold's forecast origin.

4. ACF/ADF summaries use the longest uninterrupted valid segment per city, so coverage periods are not identical across cities